

MALIGNANT TUMOR OF THE LARYNX

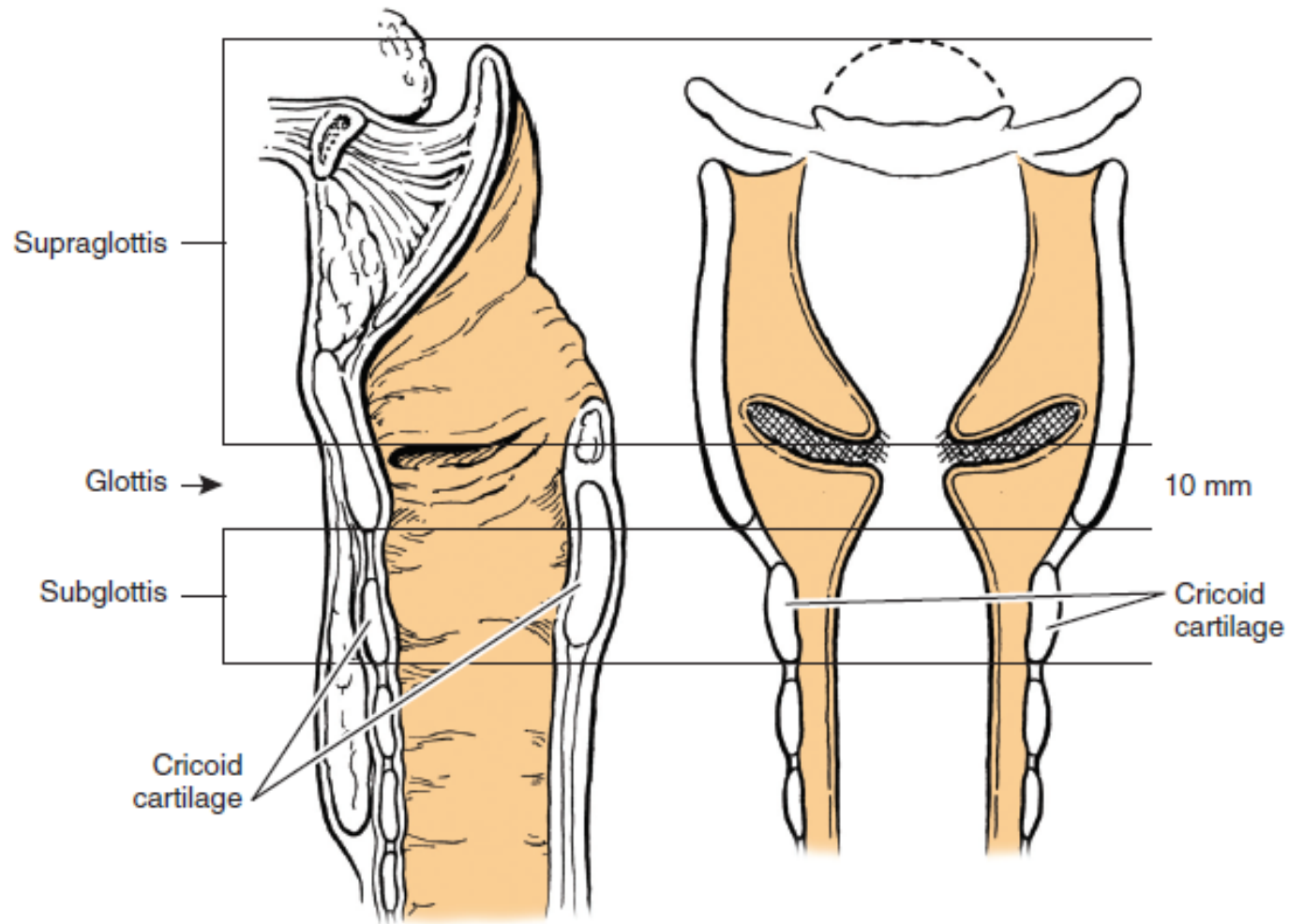
By

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ANATOMY



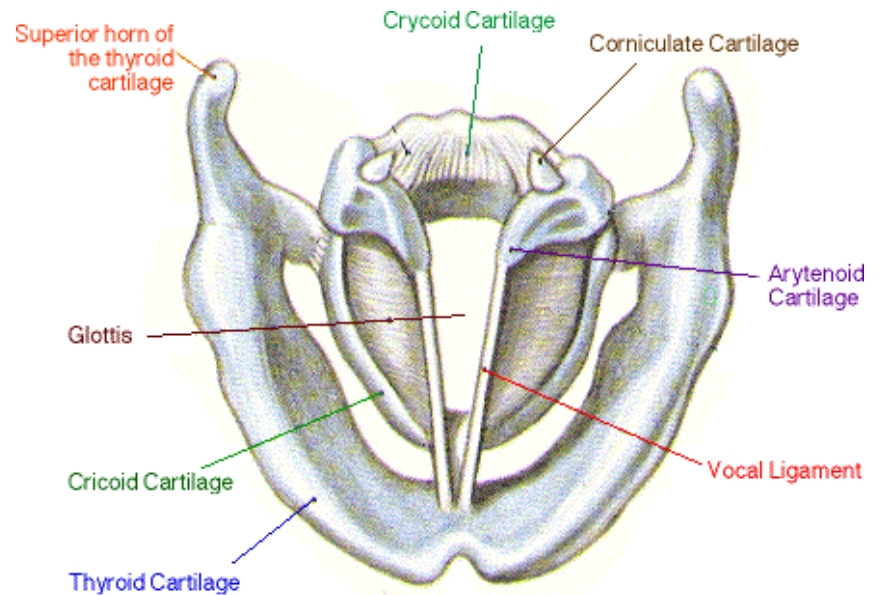
The supraglottic larynx

- epiglottis: divided into suprahyoid and infrahyoid portions
- pre-epiglottic space
- arytenoid cartilages
- aryepiglottic folds
- false vocal cords
- paraepiglottic space



The glottic larynx

- True vocal cords
- Anterior commissure
- Posterior commissure



The Larynx: viewed from above

The subglottic larynx

- extends from the inferior surface of the true vocal cords to the inferior aspect of the cricoid cartilage.



Lymphatics

- 1st-echelon lymphatics for the supraglottic larynx are subdigastric and middle anterior cervical nodes and the 2nd -echelon lymphatics are lower anterior cervical nodes
- 1st -echelon lymphatics for the sub glottic larynx are Delphian node, lower anterior cervical, paratracheal, supraclavicular nodes, and the 2nd -echelon lymphatics are mediastinal nodes.

LARYNGEAL CANCER

- It is the most common cancer of the upper aerodigestive tract.
- About 90-95% of CA larynx are SCC with various grades of differentiation .
- The rest 5-10% include verrucous ca ,spindle cell ca. and others .



SUBTYPES

- Glottic Cancer: 40-60%
- Supraglottic Cancer: 20-40%
- Subglottic Cancer: 1-10%



RISK FACTORS

- Second Head & Neck CA (highest risk).
- Smoking & Alcohol (Synergism, Site).
- GERD (Causative or Association?)
- Occupation(diesel exhaust, asbestos, organic solvents, sulfuric acid, mustard gas, certain mineral oils, metal dust, asphalt, wood dust, stone dust, mineral wool, and cement dust).
- HPV 16 &18 (Risk Factor???)
- Genetic Susceptibility.
- Diet (Duration??)



SEX & AGE INCIDENCE

- In the 1950s, the male-to-female ratio in patients with laryngeal cancer was 15:1.
- This number had changed to 5:1 by the year 2000, and the proportion of women afflicted by the disease is projected to increase in years to come.
- These changes are likely a reflection of shifts in smoking patterns, with women smoking more in recent years.
- The peak incidence is in those aged 60-70 years.

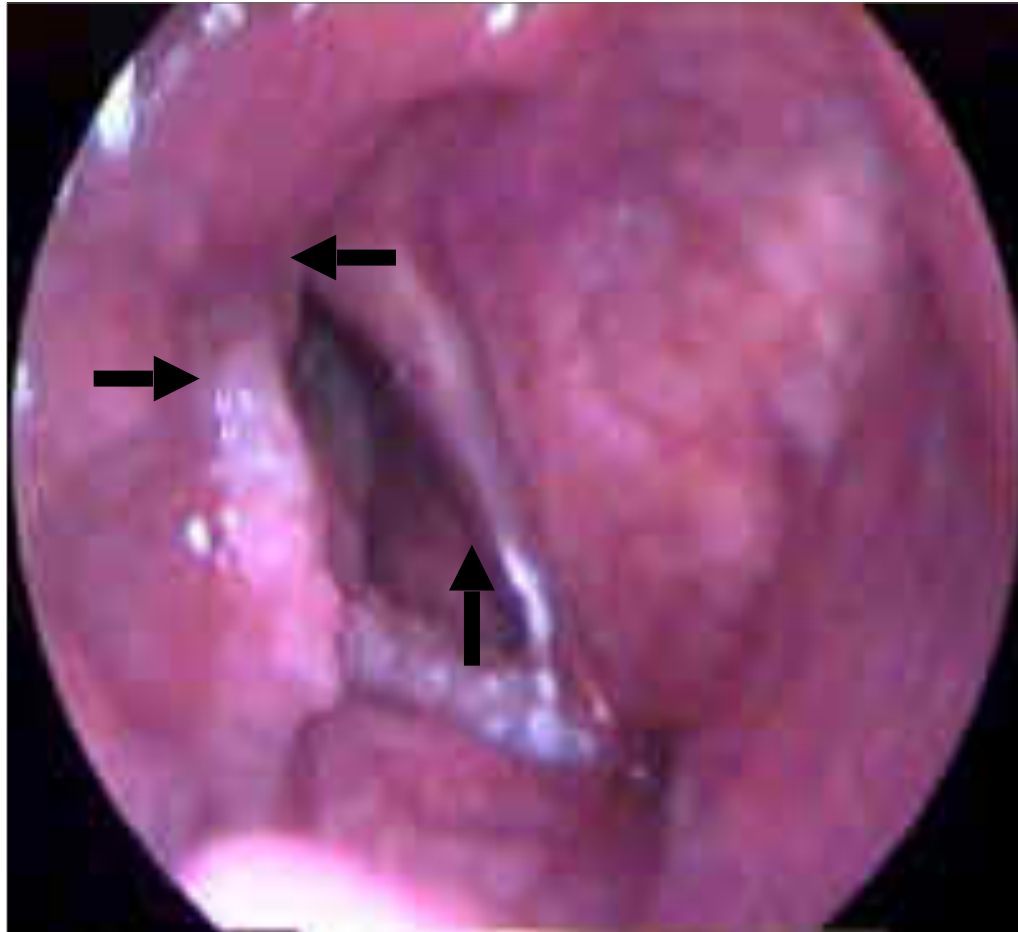


MORTALITY/MORBIDITY

- The prognosis for small laryngeal cancers that do not have lymph node metastases is good, with cure rates of 75-95%, depending on the site, the size of the tumor, and the extent of infiltration.
- Advanced disease has a worse prognosis.



STAGING



PREMALIGNANT LARYNGEAL LESIONS

WHO Classes:

- Hyperplasia
 - mild, moderate, or severe dysplasia.
 - carcinoma in situ (CIS)
-
- Malignant transformation rates of lesions without dysplasia, with mild to moderate dysplasia, and severe dysplasia/CIS of 3.8%, 10.1%, and 18.1%, respectively.
 - Malignant transformation is a multistep process that takes many years. Blackwell and colleagues²⁵ observed a 3.9-year interval between initial biopsy and progression to invasive cancer.
 - Biopsy is the gold standard for diagnosis, and adequate sampling is important, because an insufficient biopsy will result in sampling error.
 - Management: RT VS Surgery????



SUPRAGLOTTIC T STAGES

- T1 Tumor limited to one subsite of supraglottis with normal vocal cord mobility .
- T2 Tumor invades mucosa of more than one adjacent subsite of supraglottis or glottis or region outside the supraglottis (e.g., mucosa of base of tongue, vallecula, medial wall of piriform sinus) without fixation of the larynx
- T3 Tumor limited to larynx with vocal cord fixation and/or invasion of the postcricoid area, preepiglottic tissues, or paraglottic space with or without minor thyroid cartilage erosion (e.g., inner cortex)
- T4a Tumor invades through the thyroid cartilage and/or invades tissues beyond the larynx (e.g., trachea; soft tissues of neck, including deep extrinsic muscles of the tongue; strap muscles; thyroid; or esophagus)
- T4b Tumor invades prevertebral space, encases carotid artery, or invades mediastinal structures



GLOTTIC T STAGES

- T1 Tumor limited to the vocal cords; may involve anterior or posterior commissure; normal vocal cord mobility
 - T1a Tumor limited to one vocal cord
 - T1b Tumor involves both vocal cords
- T2 Tumor extends to supraglottis and/or subglottis or impairs vocal cord mobility
- T3 Tumor limited to the larynx with vocal cord fixation and/or invades paraglottic space and/or causes minor thyroid cartilage erosion (e.g., inner cortex)
- T4a Tumor invades through the thyroid cartilage and/or invades tissues beyond the larynx, (e.g., trachea; soft tissues of neck, including deep extrinsic muscles of the tongue, strap muscles, thyroid, or esophagus) .
- T4b Tumor invades prevertebral space, encases carotid artery, or invades mediastinal structures



SUBGLOTTIC T STAGES

- T1 Tumor limited to the subglottis
- T2 Tumor extends to vocal cords with normal or impaired mobility
- T3 Tumor limited to larynx with vocal cord fixation
- T4a Tumor invades cricoid or thyroid cartilage and/or invades tissues beyond larynx (e.g., trachea; soft tissues of neck, including deep extrinsic muscles of the tongue, strap muscles, thyroid, or esophagus)
- T4b Tumor invades prevertebral space, encases carotid artery, or invades mediastinal structures



STAGING- NODES

N0	No cervical lymph nodes positive
N1	Single ipsilateral lymph node $\leq 3\text{cm}$
N2a	Single ipsilateral node $> 3\text{cm}$ and $\leq 6\text{cm}$
N2b	Multiple ipsilateral lymph nodes, each $\leq 6\text{cm}$
N2c	Bilateral or contralateral lymph nodes, each $\leq 6\text{cm}$
N3	Single or multiple lymph nodes $> 6\text{cm}$



SUPRAGLOTTIC CARCINOMA

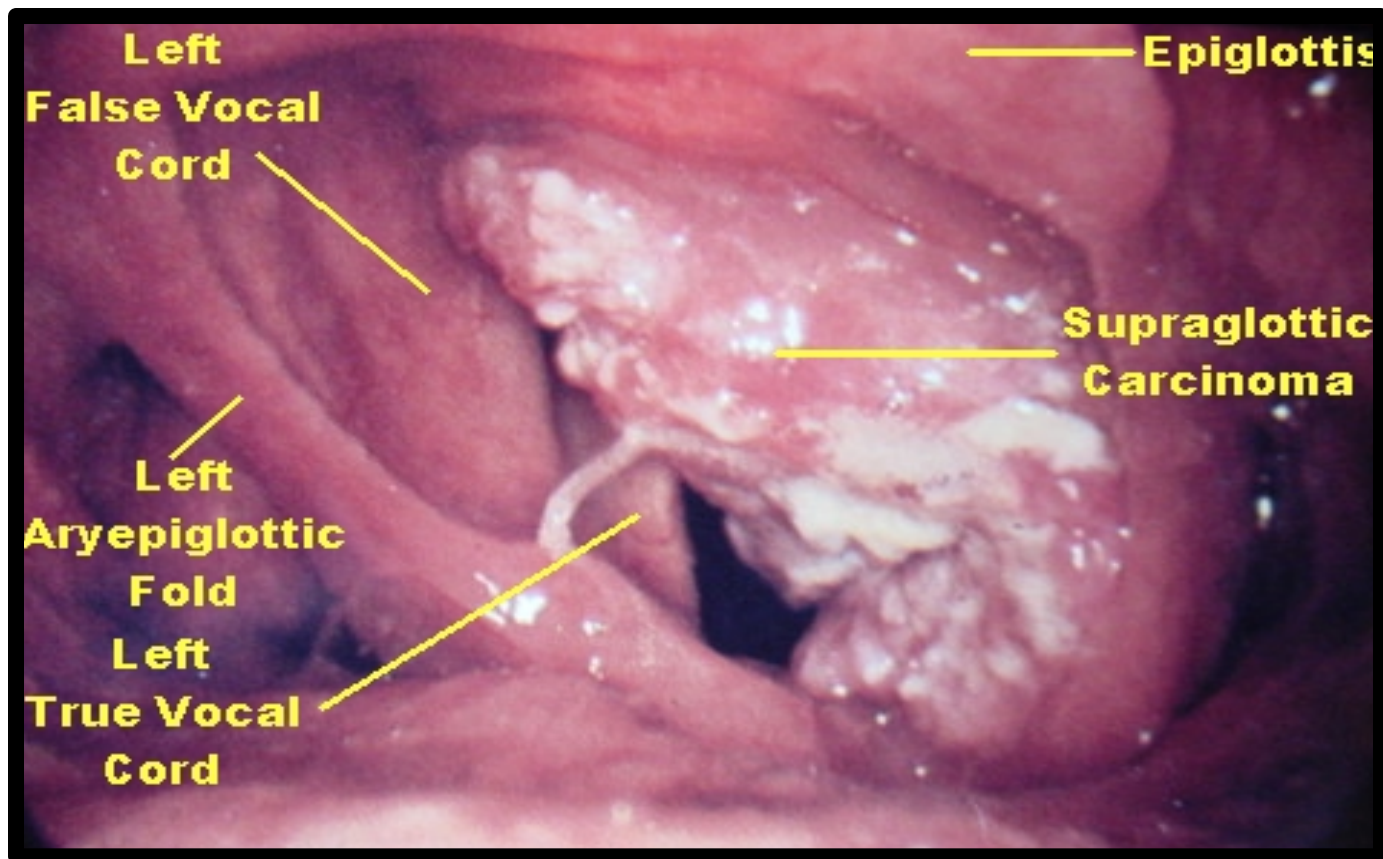
- The epiglottis is the most frequent location , these lesions are often exophytic and circumferential masses.
- Tumors of the aryepiglottic fold are typically exophytic lesions that, when detected early are confined laterally along the aryepiglottic fold.
- Advanced lesions may extend laterally to involve the adjacent wall of the pyriform sinus or medially to invade the epiglottis.



SUPRAGLOTTIC CARCINOMAS

- SCC that arise from the false vocal cords and laryngeal ventricle tend to be ulcerative and infiltrative with a limited exophytic component.
- Deep invasion by such tumors results in their access to the paraglottic space, and this may lead to fixation of the supraglottic larynx.
- Because of their close proximity, these tumors may extend inferiorly to involve the true vocal cords.





GLOTTIC CARCINOMAS

- The true vocal cords are the most common site of laryngeal carcinomas; the ratio of glottic carcinomas to supraglottic carcinomas is approximately 3:1.
- The anterior portion of the true vocal cord is the most common location of SCC with most lesions occurring along the free margin of the vocal cord.





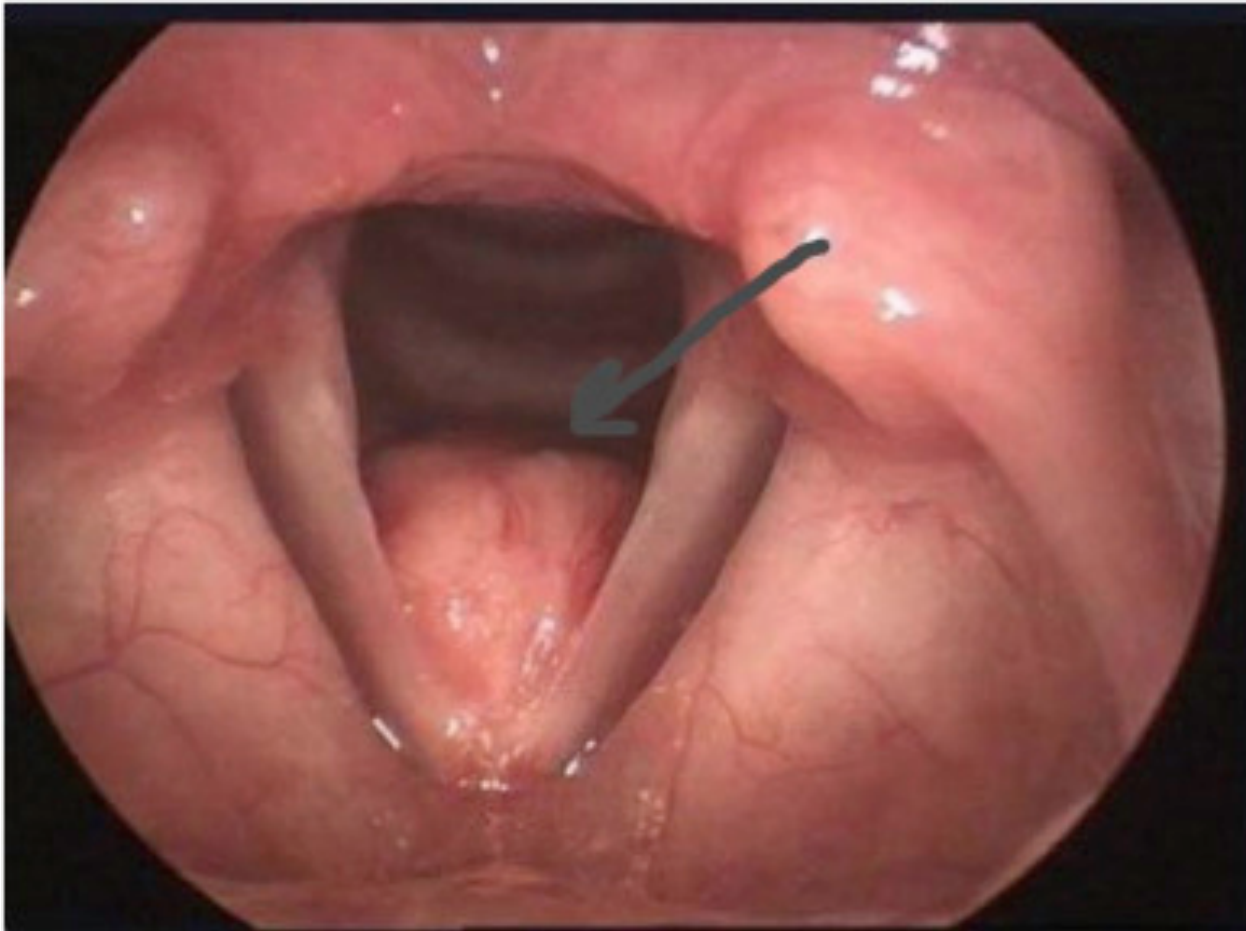
Cancer of the Lt true vocal cord



SUBGLOTTIC CARCINOMAS

- Subglottic carcinomas are rare and account for only 1% of all laryngeal carcinomas.
- When present, these lesions are characteristically circumferential and often extend to involve the undersurface of the true vocal cords .
- They have a tendency for early invasion of the cricoid cartilage and extension through the cricothyroid membrane.





Subglottic
Ca



PRESENTATION

It depends on the Site:

- ✓ Dysphonia (Cardinal Symptom)
- ✓ Dyspnea / Stridor (Advanced)
- ✓ Dysphagia
- ✓ Hemoptysis
- ✓ Throat pain
- ✓ Ear pain
- ✓ Airway compromise
- ✓ Aspiration
- ✓ Neck mass (Lymph Node)



PRESENTATION

- Patients presenting with hoarseness (More than 2 weeks) should undergo an indirect mirror exam. and/or flexible laryngoscope.
- Malignant lesions can appear as friable, fungating, ulcerative masses or be as subtle as changes in mucosal color.



EXAMINATION AND DIAGNOSIS

- ***Diagnosis will be made after consideration of:***
 1. History
 2. General examination of the patient
 3. Examination of the larynx (Laryngoscopy)
 4. Examination of the neck
 5. Radiology
 6. Clinical investigations
 7. Histological examination



WORK UP

■ Other potential modalities:

- Bronchoscopy
- Esophagoscopy
- Chest X-ray
- Liver function tests with or without US
- PET



EXAMINATION OF THE LARYNX

Examine for

- the site of the primary tumor.
- involved subsites & extension to adjacent structures.
- vocal cord mobility (normal, impaired, or fixed).
- patency of the airway.
- The base of tongue and the laryngeal skeleton are palpated to assess for extra laryngeal spread.
- The neck is palpated to assess nodal status.
- Biopsy.



EXAMINATION OF THE NECK

A palpable neck mass could be:

1. Direct spread of tumour.
2. Regional lymph nodes metastasis.
3. Enlarged thyroid lobe which suggest invasion



RADIOLOGICAL EXAMINATION

❖ CT scan .

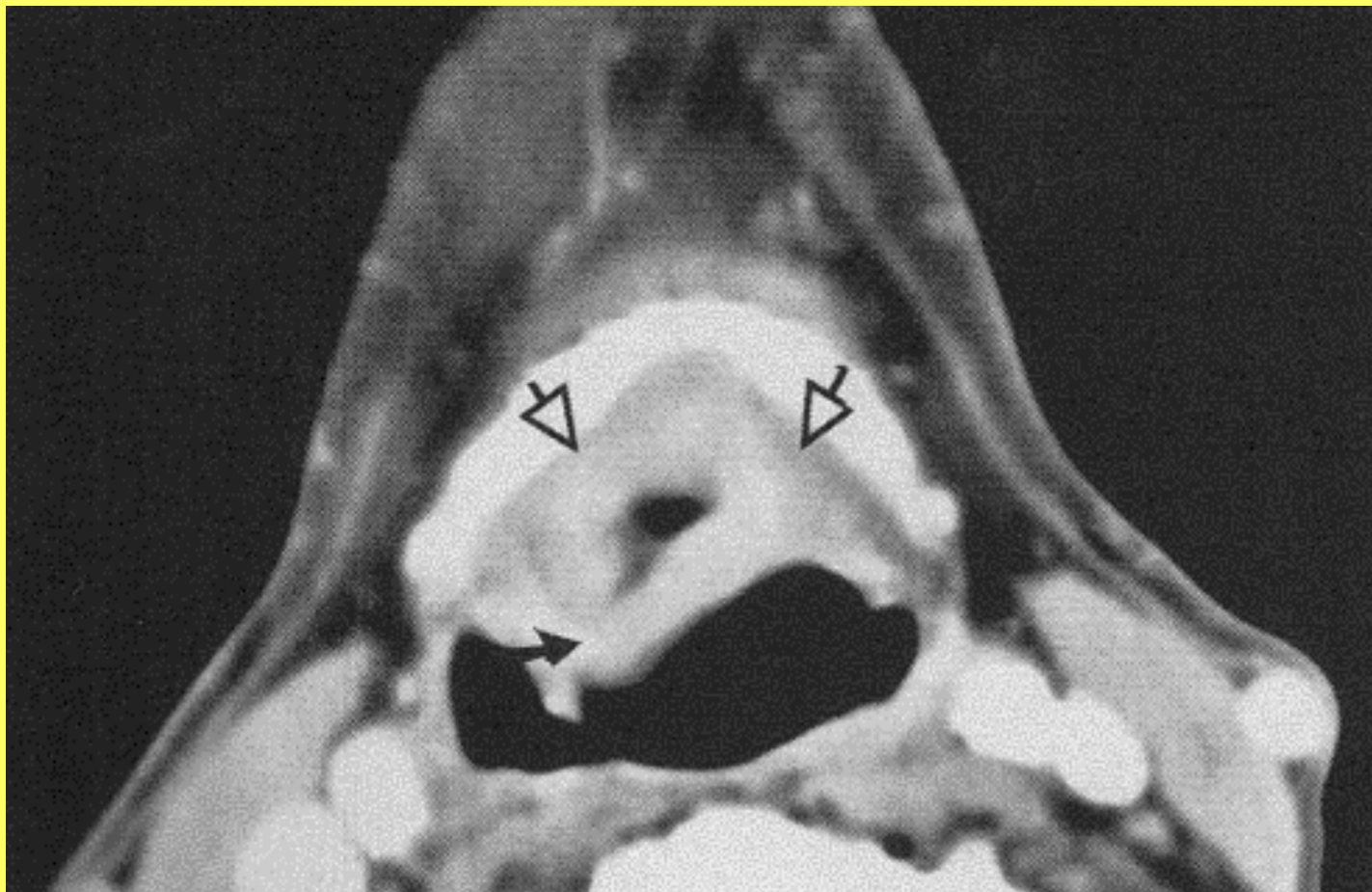
❖ MRI .

Which one????

❖ PET .



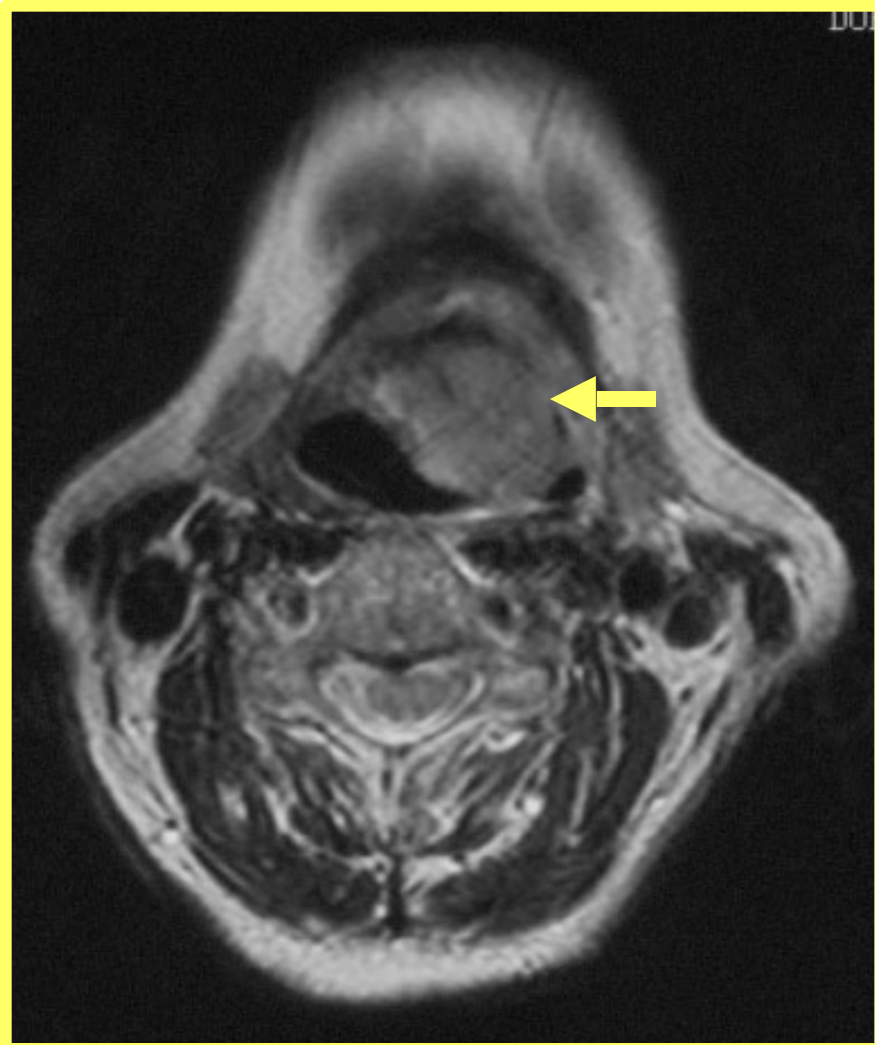
CT scan



Axial CT shows loss of pre-epiglottic fat by carcinomatous infiltration



MRI



Epiglottic tumor (laryngeal Ca. supraglottic type)



Treatment

The goals in the treatment of laryngeal cancer are to:

- 1) cure the patient;
- 2) preserve the larynx or, more correctly, preserve the function of the larynx.
- 3) minimize the morbidity of the treatment.

The most important information required for therapeutic decision making are:

- 1) the histologic diagnosis of the tumor
- 2) the site of origin of the tumor.
- 3) the stage of disease (T, N, and M stages).



TREATMENT

1. Surgery (partial , total laryngectomy +- neck dissection)
2. Chemoradiation



TREATMENT

■ XRT complications include:

- Mucositis
- Odynophagia
- Laryngeal edema
- Xerostomia
- Stricture and fibrosis
- Radionecrosis
- Hypothyroidism



FOLLOW UP

The National Comprehensive Cancer Network suggests:

- a physical examination every 1 to 3 months in the first year after treatment; every 2 to 4 months in the second year; every 4 to 6 months in the third, fourth, and fifth years; and every 6 to 12 months thereafter.
- Thyroid function tests should be checked every 6 to 12 months for patients with a history of radiation or open laryngeal surgery, because thyroid function is reduced after larynx cancer treatment that includes RT or extensive laryngeal and thyroid surgery



Reference

Cummings Otolaryngology/Head & Neck Surgery,
Sixth Edition, P 1881-1912

Thank You